EXP – 9 Three Address Code Conversion

Lex Code:

Open *9.l /mnt/c/Users/mariy/OneDrive/Desktop/Compiler/EXP-9 Save

*9.l 9.y

```
%{
#include "y.tab.h"
%}

%%
[0-9]      { yylval = yytext[0]; return NUM; }
[a-z]      { yylval = yytext[0]; return ID; }
\n         { return 0; }
.          { return yytext[0]; }
%%

```

Yacc Code:

```
%{

#include <stdio.h>
int yylex();
int yyerror(char *s);

char temp = 'A';
char GenCode(char a, char op, char b)
{
    printf("%c = %c %c %c\n", temp, a, op, b);
    return temp++;
}

%}

%token NUM ID
%left '+' '-'
%left '*' '/'

%%

S   :   E       { printf("Result = %c\n", $1); }    ;

E   :   E '+' E     { $$ = GenCode($1, '+', $3); }
    |   E '-' E     { $$ = GenCode($1, '-', $3); }
    |   E '*' E     { $$ = GenCode($1, '*', $3); }
    |   E '/' E     { $$ = GenCode($1, '/', $3); }
    |   '(' E ')'   { $$ = $2; }
    |   NUM         { $$ = $1; }
    |   ID          { $$ = $1; }
    ;

%%

int main()
{
    printf("Enter the expression: ");
    yyparse();
    return 0;
}
int yywrap()
{
    return 1;
}
int yyerror(char *s)
{
    printf("Error\n");
    return 0;
}
```

Output:

```
mariy@Venkata-HP-Touch:/mnt/c/Users/mariy/OneDrive/Desktop/Compiler/EXP-9$ lex 9.l
mariy@Venkata-HP-Touch:/mnt/c/Users/mariy/OneDrive/Desktop/Compiler/EXP-9$ yacc -d 9.y
mariy@Venkata-HP-Touch:/mnt/c/Users/mariy/OneDrive/Desktop/Compiler/EXP-9$ gcc lex.yy.c y.tab.c
mariy@Venkata-HP-Touch:/mnt/c/Users/mariy/OneDrive/Desktop/Compiler/EXP-9$ ./a.out
Enter the expression: a+b*c
A = b * c
B = a + A
Result = B
mariy@Venkata-HP-Touch:/mnt/c/Users/mariy/OneDrive/Desktop/Compiler/EXP-9$ ./a.out
Enter the expression: a+v+e*v-(e*d/8)/2+w
A = a + v
B = e * v
C = A + B
D = e * d
E = D / 8
F = E / 2
G = C - F
H = G + w
Result = H
mariy@Venkata-HP-Touch:/mnt/c/Users/mariy/OneDrive/Desktop/Compiler/EXP-9$
```